

FEDERAL UNIVERSITY LAFIA
DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATIONS 2015/2016 SESSION

COURSE TITLE: General Mathematics I

COURSE CODE: MTH 111

INSTRUCTION(S): (1) Do not write on your question paper. (2) Answer all questions in section A and any three (3) questions in section B.

TIME ALLOWED: $2\frac{1}{2}$ Hours

CREDIT UNIT(S): 3

SECTION A

1. Which of these is not true for n^{th} root of unity? (A) the roots are n in number (B) there is at least one real root (C) the sum of the roots is equal to zero (D) there is at most one real root
2. The real and imaginary parts of $\frac{2}{3+i} + \frac{3}{2+i}$ are respectively (A) $-\frac{9}{5}, \frac{4}{5}$ (B) $-\frac{4}{5}, \frac{9}{5}$ (C) $\frac{9}{5}, -\frac{4}{5}$ (D) $\frac{4}{5}, \frac{9}{5}$
3. The argument of $-\sqrt{3} + i$ is ~~(A) $\frac{\pi}{6}$~~ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $-\frac{\pi}{3}$
4. The number, $1 + i$ in polar form is ~~(A) $\sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$~~ (B) $2(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$
(C) $\sqrt{2}(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$ (D) $4(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$
5. For what values of r will the equation $x^2 + (r-1)x + 1 = 0$ have equal roots? (A) 3 and -1
(B) -1 and -3 (C) -3 and 1 (D) 1 and 3
6. Find two numbers such that their arithmetic mean is 6.5 and their geometric mean is 6. (A) 3, 6 (B) 4, 9
(C) 6, 8 (D) 4, 16
7. If the sum of infinity of the series $x + x^3 + x^5 + \dots$ is $\frac{2}{3}$, find the value(s) of x . (A) $\frac{1}{2}, 2$ (B) $\frac{1}{2}$ (C) $\frac{1}{2}, -2$
(D) 2
8. What is the n^{th} term of the sequence 9, 3, 1, ...? (A) $9\left(\frac{1}{3}\right)^{n+1}$ (B) $9\left(\frac{1}{3}\right)^n$ (C) $27\left(\frac{1}{3}\right)^{n+1}$ (D) $27\left(\frac{1}{3}\right)^n$
9. If the 9th term of an arithmetic sequence is 5 times the 5th term, find the relationship between the first term a and the common difference d . (A) $a + 2d = 0$ (B) $2a + 3d = 0$ (C) $2a + d = 0$ (D) $a + 3d = 0$
10. There are basically _____ step(s) in the proof by mathematical induction. (A) 1 (B) 3 (C) 2 (D) 4
11. The cardinality of the power set of A where $A = \{6, 7, 8\}$ is (A) 3 (B) 8 (C) 6 (D) 10
12. One of the following is DeMorgan's law of sets. (A) $(A \cup B)' = A' \cap B'$ (B) $A \cap A' = \emptyset$
(C) $(A \cup B)' = A' \cup B'$ (D) $(A \cap B)' = (A \cup B)'$
13. $\mathbb{N}$ is the set of natural numbers. Let $A = \{t: t = 2n + 1, n \in \mathbb{N}\}$ and $B = \{0 < x < 10, x \in \mathbb{N}\}$.
Find $A' \cap B'$ (A) $\{11, 13, 15, \dots\}$ (B) $\{10, 11, 12, \dots\}$ (C) $\{3, 5, 6, \dots\}$ (D) $\{10, 12, 14, \dots\}$
14. Using binomial theorem, $\left(x - \frac{1}{2x}\right)^{12} = \sum_{r=0}^{12} {}^{12}C_r x^{12-r} \left(-\frac{1}{2x}\right)^r$. For what value of r will the expression give us a constant term? (A) 3 (B) 6 (C) 5 (D) 4

15. Given the binomial expansion of $\sqrt{1-x} = 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} + \dots$. What is the value of x in the evaluation

of $\sqrt{3}$? (A) $-\frac{1}{2}$ (B) $\frac{1}{4}$ (C) -2 (D) -4

16. Find $\tan 150^\circ$. (A) $-\sqrt{3}$ (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $-\frac{1}{\sqrt{3}}$

17. Simplify $\frac{\sin \theta}{1 - \cos \theta} + \frac{\cos \theta}{1 - \sin \theta}$. (A) $\cos \theta - \sin \theta$ (B) $\sin \theta - \cos \theta$ (C) $\cos \theta$ (D) $\cos \theta + \sin \theta$

18. Solve within the range $0^\circ \leq x \leq 180^\circ$ the equation, $\sin x + 2 \sin x \cos x = 0$. (A) $x = 180^\circ, 360^\circ$

(B) $x = 0^\circ, 150^\circ$ (C) $x = 0^\circ, 120^\circ, 180^\circ$ (D) $x = 0^\circ, 150^\circ, 180^\circ, 360^\circ$

19. If $\theta = 225^\circ$, what is θ in radians? (A) $\frac{5\pi}{6}$ (B) $\frac{5\pi}{4}$ (C) $\frac{4\pi}{5}$

20. If $\theta = \frac{3\pi}{4}$ in radians, what is θ in degrees? (A) 135° (B) 125° (C) 45° (D) 220°

SECTION B

1. (a) In an examination, 36 boys offered French, 32 offered German and 31 offered Spanish, 9 offered French and German, 12 offered German and Spanish, and 11 offered French and Spanish. There were 4 boys who offered all three languages and 9 boys offered none of the languages. Let F, G and S represent the set of boys who offered French, German and Spanish respectively.

(i) Represent the information on a Venn diagram and find how many boys are there altogether.

(ii) How many boys offered at least one of the languages?

(b) Find the term independent of x in the binomial expansion of $(2 + \frac{1}{x})^{10}$

2. (a) The 3rd term of an A.P. is 7 and the sum of the first ten terms of the A.P. is 120. Find the A.P.

(b) Given that α and β are the roots of $x^2 + x - 1 = 0$. Form a quadratic equation of the roots $\alpha + 1$ and $\beta + 1$.

(a) Find the roots of $z^3 = 8$, where z is a complex number.

(b) If $A = \{2, -3\}$, $B = \{x: x \text{ is the root of } x^2 - x - 2 = 0\}$. Show that A and B are not disjoint.

4. Solve $2\cos^2 x - \sin x = 1$ for $x \in [0, 2\pi]$

(b) Given the set $A = \{x, y, z\}$, Find the power set of A .

(a) Prove that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$ by mathematical induction.

(b) Use the discriminant to determine the nature of the roots of $3y^2 - 22y + 5 = 0$

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1. What will be added to $2x^2 + 5x$ to make it a perfect square (A) $\frac{25}{8}$ (B) $\frac{10}{4}$ (C) $\frac{25}{16}$ (D) $\frac{25}{4}$
2. Given $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, evaluate $f(A) = A^2 - 4A + 3I$. (A) $\begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (C) $\begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$ (D) $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$
3. What are the possible values of p for which $4x^2 = 2px - 25$ has equal roots? (A) ± 10 (B) ± 5 (C) $5, -10$ (D) $-5, 10$
4. Solve the equation $\cos x + \sin x = \frac{1}{\cos x - \sin x}$ for values of x such that $0 \leq x < 2\pi$. (A) $0, \pi$ (B) $\frac{\pi}{3}, \frac{2\pi}{3}$ (C) $0, \frac{\pi}{3}$ (D) $\frac{\pi}{2}, \frac{3\pi}{2}$
5. Three consecutive terms in a geometric sequence are $n-2, n$ and $n+3$. What is the common ratio of the sequence? (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{2}$
6. Using the first two terms of the series in the binomial expansion of $\sqrt{1-x} = 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} + \dots$, $\sqrt{3}$ can be approximated as..... (A) $\frac{11}{6}$ (B) $\frac{15}{8}$ (C) $\frac{7}{4}$ (D) $\frac{4}{3}$
7. Using the binomial theorem, $(2x + \frac{1}{3x})^{10} = \sum_{r=0}^{10} {}^{10}C_r x^{10-r} (\frac{1}{3x})^r$. For what value of r will the expression give a constant term? (A) 4 (B) 7 (C) 2 (D) 5
8. If $R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, what is R^{-1} ? (A) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (C) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (D) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Use the following information provided below to answer questions 9 - 12.

Out of 90 travelers interviewed, 7 travelled by bus and train only, 3 by train and car only, 8 travelled by three means of transport. The number, x , of travelers who travelled by bus only was equal to the number who travelled by bus and car only. If 47 people travelled by bus and 30 by train.

9. What is the value of x ? (A) 18 (B) 17 (C) 16 (D) 15
10. How many people travelled by exactly two means? (A) 29 (B) 26 (C) 27 (D) 28
11. How many people travelled by car? (A) 55 (B) 56 (C) 57 (D) 58
12. What percentage of the people travelled by bus or train but not car? (A) 56.7% (B) 38.9% (C) 7.8% (D) 35.5%
13. $\mathbb{N}$ is the set of natural numbers. Let $A = \{t: t = 2n + 1, n \in \mathbb{N}\}$ and $B = \{0 < x \leq 10, x \in \mathbb{N}\}$. Find $A' \cap B$ (A) $\{11, 13, 15, \dots\}$ (B) $\{10, 12, 14, \dots\}$ (C) $\{3, 5, 6, \dots\}$ (D) $\{10, 11, 12, \dots\}$
14. If the 10th term of an A.P. is double the 3rd term, what is the relationship between the first term a and the common difference d . (A) $a + 5d = 0$ (B) $2a + 11d = 0$ (C) $a - 5d = 0$ (D) $2a - 11d = 0$
15. The minor of the entry, a_{23} of a given matrix is -14 . What is the value of the cofactor of the entry? (A) 15 (B) -14 (C) 13 (D) 14
16. If $\cos \theta = \frac{12}{13}$, what is the value of $\sin \theta$? (A) $\frac{13}{5}$ (B) $\frac{5}{13}$ (C) $\frac{13}{6}$ (D) $\frac{5}{13}$
17. Given two matrices S and T with orders (4×5) and (5×6) respectively. The matrix ST has order (A) (5×5) (B) (4×5) (C) (4×6) (D) (5×6)
18. The determinant of $P = \begin{pmatrix} \cos 30^\circ & \sin 30^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix}$ is (A) $\frac{\sqrt{3}}{2}$ (B) 0 (C) $\frac{\sqrt{3}}{4}$ (D) $\frac{1}{2}$
19. Find the values of x which satisfy the equation $\frac{2}{x+2} + \frac{1}{x} = 1$. (A) -1 and 2 (B) 1 and 2 (C) -1 and -2 (D) 1 and -2
20. A matrix A is skew symmetric if..... (A) $A' = -A$ (B) $A = A^T$ (C) $A = -A^T$ (D) $A^T = -A$

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DEPARTMENT OF MATHEMATICS
FEDERAL UNIVERSITY, LAFIA

MTH111 2018 C.A Test

Time: 1hr

13 1/2
30

Instructions: Answer all Questions

- If $x - 2$, $x - 1$, and $3x - 5$ are the three consecutive terms of a G.P. find two possible values of the common ratios. (a) 2 and 1 (b) 2 and 3 (c) 2 and 1 (d) $\frac{3}{2}$ and 3 (e) 3 and 1
- The n th term of the sequence $4, 2, 1, \frac{1}{2}, \dots$ is (a) $8\left(\frac{1}{2}\right)^n$ (b) 2^{n-1} (c) $8\left(\frac{1}{2}\right)^{-n}$ (d) 2^{2n+1} (e) 2^{n+1}
- If α and β are the roots of $x^2 - 7x + 10 = 0$. Find the value of $\alpha^3 + \beta^3$. (a) 133 (b) $\frac{1}{3}$ (c) 123 (d) 273 (e) 132
- If there are 512 possible subsets in set X , compute the cardinality of set X (a) 7 (b) 9 (c) 6 (d) 8 (e) 5. ??
- If $A = \{5, 10, 15, 20, 25, 30, 35\}$ and $B = \{4, 6, 8, 10, 20, 30\}$, find $A - B$. (a) $\{4, 6, 8\}$ (b) $\{5, 15, 25, 30\}$ (c) $\{4, 6, 8, 10, 20, 30\}$ (d) $\{5, 15, 25, 35\}$ (e) non of the above.
- Find the sum of $\frac{3}{4}, \frac{2}{3}, \frac{7}{12}, \dots$ up to the 19th terms. (a) $\frac{57}{2}$ (b) 57 (c) $\frac{57}{4}$ (d) 0 (e) 12
- Find the number of ways of permutation of the letters of the word WELL such that the 2Ls will always be apart. (a) 6ways (b) 24ways (c) 8ways (d) 30 (e) none of the above.
- Write the term in x^2 in the binomial expansion of $(x - \frac{1}{2x})^{12}$. (a) $-\frac{1}{8}x^2$ (b) $-\frac{99}{4}x^2$ (c) $\frac{45}{4}x^2$ (d) $-\frac{45}{8}x^2$ (e) $-\frac{99}{8}x^2$
- Out of 5 women and 7 men, a committee consisting of 2 women and 3 men is to be formed. In how many ways can this be done if one particular man must be in the committee? (a) 31 5ways (b) 350ways (c) 150ways (d) 840ways (e) 21ways
- Convert 135° to an angle in radian (a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) π (d) $\frac{\pi}{2}$ (e) $\frac{3\pi}{4}$
- The sum of three numbers in an A.P is 15 and the sum of the squares of its first and third term is 59. Find the number. (a) 2, 5, 8 (b) 1, 4, 10 (c) 3, 3, 9 (d) 2, 3, 10 (e) 3, 5, 7
- If $\tan\theta = 1$ and $\sin\theta$ is negative. Find $\cos\theta$. (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $-\sqrt{2}$ (d) $\sqrt{2}$ (e) $-\frac{\sqrt{2}}{2}$
- Determine the nature of the roots of $4x^2 + 2x + 3 = 0$. (a) Real and distinct (b) stationary roots (c) equal roots (d) real and equal (e) complex roots.
- Which of the six trigonometrical ratios are positive for 960° ? (a) sine and tangent (b) sine and cosine (c) tangent and cotangent (d) secant and sine (e) cosecant and cotangent.
- How many terms of the following series may be taken so that their sum is 66? $-9, -6, 3, \dots$? (a) -4 (b) 5 (c) 11 (d) 10 (e) 12
- Write 24, a composite number as a product of prime factors (a) $2 \times 3 \times 4$ (b) $1 \times 2 \times 3 \times 4$ (c) $2 \times 2 \times 2 \times 3$ (d) $1 \times 2 \times 2 \times 2 \times 3$ (e) 4×6
- Given that $\sin A = \frac{3}{5}$ and $\cos B = \frac{8}{17}$, where A and B are acute angles. Find $\cos(A + B)$. (a) $\frac{6468}{7225}$ (b) $\frac{77}{85}$ (c) $\frac{468}{7225}$ (d) $\frac{-13}{85}$ (e) $\frac{81}{93}$
- If α and β are the roots of $2x^2 + 3x + 7 = 0$. Find the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$. (a) $\frac{19}{15}$ (b) $-\frac{74}{14}$ (c) $\frac{74}{14}$ (d) $-\frac{19}{14}$ (e) $\frac{18}{13}$
- For what values of m will the equation $(m + 1)x^2 + 2(m + 3)x + (2m + 3) = 0$ have equal roots? (a) 1 or -1 (b) 5 or -3 (c) 1 or 1 (d) -2 or 3 (e) 3 or 4

$$\alpha^3 + \beta^3 = -\frac{10}{9} \quad \frac{c}{a} = \frac{10}{1}$$

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

$$(\alpha + \beta)^3 = 2 \times 10$$

$$6 \times 4$$

$$\begin{array}{r} 2 \overline{) 24} \\ 4 \\ \hline 2 \overline{) 12} \\ 6 \\ \hline 2 \overline{) 6} \\ 3 \\ \hline \end{array}$$

**FEDERAL UNIVERSITY LAFIA
DEPARTMENT OF MATHEMATICS**

FIRST SEMESTER EXAMINATIONS 2016/2017 SESSION

COURSE TITLE: General Mathematics I

COURSE CODE: MTH 111

INSTRUCTION(S): (1) Do not write on your question paper. (2) Answer all questions.

TIME ALLOWED: $2\frac{1}{2}$ Hours

CREDIT UNIT(S): 3

1. Find the determinant of the matrix, $A = \begin{pmatrix} 2 & -6 & 10 \\ 4 & 0 & -2 \\ 3 & -2 & 8 \end{pmatrix}$
(A) 0 (B) -316 (C) -26 (D) 208 (E) 120
2. Calculate the argument of $\cos \frac{1}{4}\pi + i \sin \frac{1}{4}\pi$. (A) π (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{3\pi}{4}$ (E) $\frac{3\pi}{2}$
3. The cardinality of the power set of A where $A = \{b, c, d, e\}$ is (A) 4 (B) 10 (C) 8 (D) 16 (E) 32
4. Using the first two terms of series in the binomial expansion of $\sqrt{1-x} = 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} + \dots$.
Evaluate $\sqrt{3}$. (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) -2 (D) $\frac{1}{4}$ (E) $\frac{7}{4}$
5. Solve within the range $0^\circ \leq x \leq 180^\circ$ the equation, $\sin x + 2 \sin x \cos x = 0$. (A) $x = 90^\circ, 180^\circ$
(B) $x = 0^\circ, 150^\circ$ (C) $x = 180^\circ, 360^\circ$ (D) $x = 0^\circ, 150^\circ, 180^\circ, 360^\circ$ (E) $x = 0^\circ, 120^\circ, 180^\circ$
6. Find the modulus of $\cos \theta + i \sin \theta$. (A) -i (B) -1 (C) i (D) 1 (E) 2i
7. The complex number $\sqrt{3} - i$ has modulus as (A) -3 (B) 4 (C) 2 (D) $\frac{\pi}{6}$ (E) $\frac{\pi}{4}$
8. If $\theta = 225^\circ$, what is θ in radians? (A) $\frac{5\pi}{6}$ (B) $\frac{5\pi}{4}$ (C) $\frac{4\pi}{5}$ (D) $\frac{6\pi}{5}$ (E) $\frac{4\pi}{3}$
9. If the sum of infinity of the series $x + x^3 + x^5 + \dots$ is $\frac{2}{3}$, find the value(s) of x. (A) $\frac{1}{2}, -2$ (B) $\frac{1}{2}$
(C) $\frac{1}{2}, 2$ (D) 2 (E) 3
10. A matrix A is skew symmetric if (A) $A' = -A$ (B) $A = A^T$ (C) $A = -A^T$ (D) $A^T = -A^T$ (E) $A = -A$
11. If $\frac{2+9i}{5-2i} = x + iy$, where x and y are real. What are the values of x and y?
(A) $(-\frac{9}{2}, \frac{2}{5})$ (B) $(-\frac{8}{29}, \frac{49}{29})$ (C) $(\frac{49}{29}, \frac{8}{29})$ (D) $(\frac{2}{5}, -\frac{9}{2})$ (E) $(\frac{49}{29}, \frac{8}{29})$
12. Find $\tan 150^\circ$. (A) $-\sqrt{3}$ (B) $\sqrt{3}$ (C) $-\frac{1}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{3}}$ (E) $\frac{\sqrt{3}}{2}$
13. There are basically _____ step(s) in the proof by mathematical induction. (A) 1 (B) 4 (C) 3 (D) 2 (E) 5
14. If $z = 3 + 5i$, find $z - \bar{z}$. (A) 6 (B) 25 (C) 34 (D) $-4 + 15i$ (E) $10i$
15. For what values of x is $2 \cos^2 x - \sin x = 1$ for $x \in [0, 2\pi]$. (A) $\frac{\pi}{3}, \frac{2\pi}{3}, \pi$ (B) $\frac{\pi}{6}, \frac{2\pi}{6}, \frac{3\pi}{4}$
(C) $0, \frac{\pi}{3}, \frac{2\pi}{3}$ (D) $0, \pi, \frac{3\pi}{2}$ (E) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$
16. Out of 400 students in Faculty of Science in the Federal University Lafia, 100 took MTH 111 as elective and 150 took PHY 111 as elective. If 75 took both MTH 111 and PHY 111 as electives, how many students took neither MTH 111 nor PHY 111?
(A) 325 (B) 25 (C) 250 (D) 225 (E) 225
17. The real and imaginary parts of $\frac{7-i}{1+7i}$ are respectively. (A) $\frac{7}{25}, -1$ (B) $\frac{14}{50}, -\frac{48}{50}$ (C) 0, -1
(D) $\frac{4}{5}, \frac{9}{5}$ (E) $\frac{9}{5}, -\frac{4}{5}$

18. Express $(3 - 8i)(5 + 7i)$ in the $x + iy$ form.
 (D) $-71 + 66i$ (E) $-41 - 24i$
19. Solve the equation $\cos x + \sin x = \frac{1}{\cos x - \sin x}$ for values of x such that $0 \leq x < 2\pi$.
 (A) $0, \pi$ (B) $\frac{\pi}{3}, \frac{2\pi}{3}$ (C) $0, \frac{\pi}{3}$ (D) $\frac{\pi}{3}, \pi$ (E) $\frac{\pi}{2}, \frac{3\pi}{2}$
20. Let $P = \{a, b, c, \dots, h, i\}$, $Q = \{b, d, f, h\}$, $R = \{a, c, e, g, i\}$, $S = \{c, d, e\}$ and $T = \{c, e\}$.
 If X and Q are disjoint, find X .
 (A) R and T (B) T (C) S and T (D) S (E) R
21. The determinant of $P = \begin{pmatrix} \cos 30^\circ & \sin 30^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix}$ is (A) $\frac{1}{2}$ (B) 0 (C) $\frac{\sqrt{3}}{4}$ (D) $\frac{\sqrt{3}}{2}$ (E) $\frac{3}{2}$
22. Find two numbers such that their arithmetic mean is 6.5 and their geometric mean is 6.
 (A) 3, 6 (B) 4, 16 (C) 4, 9 (D) 6, 8 (E) 5, 7
23. If $\cos \theta = \frac{12}{13}$, what is the value of $\sin \theta$? (A) $\frac{13}{5}$ (B) $\frac{6}{13}$ (C) $\frac{13}{6}$ (D) $\frac{5}{13}$ (E) $\frac{5}{12}$
24. One of the following is DeMorgan's law of sets. (A) $(A \cup B)' = (A \cup B)'$ (B) $A \cap A' = \emptyset$
 (C) $(A \cup B)' = A' \cup B'$ (D) $A' \cup A = \xi$ (E) $(A \cup B)' = A' \cap B'$
25. Given that $\begin{pmatrix} 3x & 5 \\ -1 & y \end{pmatrix} = \begin{pmatrix} 9 & 5 \\ -1 & -10 \end{pmatrix}$, Find the value of $3x - 2y$?
 (A) -27 (B) -28 (C) 13 (D) 10 (E) 29
26. If $\theta = \frac{7\pi}{6}$ in radians, what is θ in degrees? (A) 45° (B) 125° (C) 220° (D) 210° (E) 205°
27. Given two matrices S and T with orders (4×5) and (5×6) respectively. The matrix ST has order
 (A) (5×5) (B) (4×5) (C) (4×6) (D) (5×6) (E) (5×4)
28. If the 9th term of an arithmetic sequence is 5 times the 5th term, find the relationship between the first term, a and the common difference, d .
 (A) $a + 2d = 0$ (B) $a + 3d = 0$ (C) $2a + 3d = 0$ (D) $2a + d = 0$ (E) $3a + d = 0$
29. If $A = \{1, 2, a, b, c, d\}$, $B = \{2, 3, a, b, c, d, e, f\}$, $C = \{2, 4, c, d, h\}$ and $D = \{1, a, 2, b, 3, c\}$. Obtain $(A \cap B) \cap (C \cup D)$
 (A) $\{2, a, b, c, d\}$ (B) $\{1, a, b, 3\}$ (C) $\{2, 1, a, b, c\}$ (D) $\{b, 4, h\}$ (E) $\{1, b\}$
30. The value of $\tan(-15^\circ)$ in surd form is.....
 (A) $\sqrt{3} - 2$ (B) $2 - \sqrt{3}$ (C) $\sqrt{3} + 2$ (D) $2 + \sqrt{3}$ (E) $\sqrt{3} + 3$
31. The minor of the entry, a_{23} of a given matrix is -14 . What is the value of the cofactor of the entry?
 (A) -14 (B) 14 (C) 13 (D) 15 (E) -13
32. What is the n th term of the sequence 9, 3, 1, ...?
 (A) $9\left(\frac{1}{3}\right)^{n+1}$ (B) $27\left(\frac{1}{3}\right)^{n+1}$ (C) $27\left(\frac{1}{3}\right)^n$ (D) $9\left(\frac{1}{3}\right)^n$ (E) $3\left(\frac{1}{3}\right)^{n-1}$
33. Find the coefficient of x^4 in the expansion of $\left(x^4 + \frac{1}{x^3}\right)^{15}$
 (A) 15 (B) ${}^{15}C_3$ (C) ${}^{12}C_5$ (D) ${}^{12}C_8$ (E) 120
34. If $\cos \theta = \frac{\sqrt{3}}{2}$ and $\tan \theta$ is negative, what is $\sin \theta$? (A) $\frac{\sqrt{2}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $-\frac{\sqrt{3}}{2}$ (E) $-\frac{1}{2}$
35. If $R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, what is R^{-1} ? (A) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (C) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (D) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ (E) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
36. For what values of r will the equation $x^2 + (r - 1)x + 1 = 0$ have equal roots?
 (A) 1 and 3 (B) -1 and -3 (C) -3 and 1 (D) 3 and -1 (E) 2 and 3

37. In a class of 40 students, 24 play football, 18 play hockey and 6 do not play any game. How many play both? (A) 3 (B) 2 (C) 8 (D) 1 (E) 5
38. Three consecutive terms in a geometric sequence are $n-2$, n and $n+3$. What is the common ratio of the sequence? (A) $\frac{1}{3}$ (B) $\frac{2}{2}$ (C) $\frac{1}{3}$ (D) $\frac{3}{2}$ (E) $\frac{4}{3}$
39. Given $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, evaluate $f(A) = A^2 - 4A + 3I$. (A) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (B) $\begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix}$
(C) $\begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$ (D) $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (E) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
40. The number, $1+i$ in polar form is (A) $\sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$ (B) $2(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$
(C) $\sqrt{2}(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$ (D) $4(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$ (E) $\sqrt{2}(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$
41. A diagonal matrix whose diagonal elements are one (1) is called
(A) Triangular matrix (B) Unit Matrix (C) Diagonal Matrix (D) Null Matrix (E) Square Matrix
42. The argument of $-\sqrt{3} + i$ is (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{5\pi}{6}$ (D) $-\frac{\pi}{3}$ (E) $\frac{3\pi}{4}$
43. Find the number of ways of selecting 3 prefects from a class of 18 students.
(A) 1253 (B) 4896 (C) 18 (D) 816 (E) 81
44. Evaluate $\sqrt{-49}$. (A) $12i$ (B) 9 (C) $-7i$ (D) $2\sqrt{2}i$ (E) $7i$
45. Given that $A = \begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$, find $A^T \cdot A$.
(A) $\begin{pmatrix} 4 & 2 \\ 0 & -5 \end{pmatrix}$ (B) $\begin{pmatrix} 21 & 2 \\ -2 & 1 \end{pmatrix}$ (C) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (D) $\begin{pmatrix} 20 & -2 \\ -4 & 1 \end{pmatrix}$ (E) $\begin{pmatrix} 25 & -2 \\ -2 & 3 \end{pmatrix}$
46. If α and β are the roots of $x^2 + px + q = 0$ with $\alpha - \beta = 5$ and $\frac{\alpha}{\beta} = \frac{7}{2}$. Find p and q .
(A) $p = 3, q = 0$ (B) $p = 1, q = 10$ (C) $p = -3, q = 10$ (D) $p = -9, q = 14$ (E) $p = 14, q = -9$

$$\begin{pmatrix} 2 & -6 & 10 & 2 & -6 \\ 1 & 0 & -2 & 1 & 0 \\ 3 & -2 & 8 & 3 & -2 \end{pmatrix}$$

$$(2 \times 0 \times 8) + (-6 \times -2 \times 3) + 10 \times 4 \times -2 - 10 \times 0 \times 3 + 2 \times -2 \times -2 + 1 \times 4 \times 8$$

$$36 - 80 - 80 + 8 - 192$$

$$-44 + 184$$

$$15C_8$$

$$2 \begin{pmatrix} 0 & -2 \\ -2 & 8 \end{pmatrix} + 6 \begin{pmatrix} 2 & -2 \\ 3 & 8 \end{pmatrix} + 10 \begin{pmatrix} 4 & 0 \\ 3 & -2 \end{pmatrix}$$

$$20 \begin{pmatrix} 2 & 6 \\ 3 & 8 \end{pmatrix} + 10 \begin{pmatrix} -8 & -5 \end{pmatrix}$$

$$6 \begin{pmatrix} 3 & 8 \end{pmatrix} + (-10)$$

$$nC_r a^{n-r} b^r$$

$$5C_3 \times 2^{15-3}$$

$$10C_3$$

$$2 \begin{pmatrix} 0 & 4 \end{pmatrix}$$

DEPARTMENT OF MATHEMATICS
FEDERAL UNIVERSITY, LAFIA
FIRST SEMESTER 2017/2018 EXAMINATION

Course Code: MTH 111 General Mathematics I (3 Units)

Instructions: Answer All Questions

Time: 2:30min

1. If the 21st and 22nd terms in the expansion of $(1+x)^{44}$ are equal. Find the value of x
(A) 6 (B) 10 (C) $\frac{3}{2}$ (D) $\frac{5}{2}$ (E) $\frac{7}{8}$
 2. In a geometric progression, the sum of the 2nd and 3rd term is 9 and the 7th term is eight times the 4th term. Find the common ratio. (A) -3 (B) $\frac{2}{3}$ (C) 3 (D) $\frac{3}{2}$ (E) 2
 3. Find the range of values of p if the equation $px^2 - 4(p-1)x + 9 = 0$ has real roots (A) $p \leq 4$ or $p \geq 2$
(B) $p \leq \frac{1}{2}$ or $p \geq 2$ (C) $p \leq \frac{1}{2}$ or $p \geq 3$ (D) $p \leq -\frac{1}{2}$ or $p \geq 4$ (E) $p \leq \frac{1}{4}$ or $p \geq 4$
 4. Given that $\begin{pmatrix} 3x & 5 \\ -1 & y \end{pmatrix} = \begin{pmatrix} 9 & 7 \\ 8 & -10 \end{pmatrix}$, what is the value of $3x + 2y^2$? (A) -10 (B) 29 (C) -209 (D) -29
(E) 209
 5. If $\sin x = \frac{1}{2}$ and x is obtuse, what is $\cot x$? (A) $\sqrt{3}$ (B) $\sqrt{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) 3 (E) $-\sqrt{3}$
- The first and the last term of an arithmetic progression are $7\frac{1}{2}$ and 93 respectively. If the common difference is $4\frac{1}{2}$. Find the sum of the series. (A) 1005 (B) 15 (C) 10005 (D) 10015 (E) 105
7. The number of radians in one complete revolution is (A) 3π (B) 2π (C) π (D) $\frac{\pi}{2}$ (E) 360°
 8. If $A = \{5, 10, 15, 20, 25, 30, 35\}$ and $B = \{4, 6, 8, 10, 20, 30\}$, find $B - A$. (A) $\{4, 6, 8, 10, 20, 30\}$
(B) $\{5, 15, 25, 30\}$ (C) $\{4, 6, 8\}$ (D) $\{5, 15\}$ (E) non of the above.
 9. If there are 128 possible subsets in set X, compute the cardinality of set X. (A) -8 (B) 8 (C) 6 (D) 7 (E) 5.
 10. Solve $2\cos^2 x - \sin x = 1$. (A) $\sin x = \frac{\sqrt{3}}{2}$ or -1 (B) $\sin x = -\frac{1}{2}$ or 1 (C) $\sin x = 1$ or -1
(D) $\sin x = -\frac{1}{2}$ or $\frac{1}{2}$ (E) $\sin x = \frac{1}{2}$ or -1
 11. Without solving the quadratic equation, define the nature of the roots of $4x^2 + x - 3 = 0$
(A) real and distinct (B) equal roots (C) not real roots (D) $-1, \frac{3}{4}$ (E) $2, \frac{3}{4}$
 12. The compound angle identity $\cos(A+B)$ is given by (A) $\sin A \sin B + \cos A \cos B$
(B) $\cos A \cos B - \sin A \sin B$ (C) $\sin A \sin B - \cos A \cos B$ (D) $\sin A \cos B + \sin B \cos A$
(E) $\cos A \cos B + \sin A \sin B$
 13. If $\cos \theta = \frac{\sqrt{3}}{2}$ and $\tan \theta$ is negative, what is $\sin \theta$? (A) $\sin \theta = \frac{1}{2}$ (B) $\sin \theta = -\frac{\sqrt{3}}{2}$ (C) $\sin \theta = -\frac{1}{2}$
(D) $\sin \theta = \frac{3}{2}$ (E) $\sin \theta = \frac{2}{3}$

14. The 2nd term of a geometry series is 24, the 5th term is 81. Find the first term. (A) 14 (B) $\frac{3}{2}$ (C) 12 (D) 10 (E) -12
 $24 = ar$ $81 = ar^4$ $\frac{81}{24} = r^3$ $r = \frac{3}{2}$ $a = 12$
15. Out of 6 women and 8 men, a committee consisting of two women and three men is to be formed. In how many ways can this be done if one particular man must be on the committee? (A) 840 ways (B) 525 ways (C) 420 ways (D) 300 ways (E) 315 ways
16. The modulus of $\frac{1-i}{1+i}$ is (A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) $\sqrt{3}$ (D) 2 (E) -2
17. Simplify $(1 - \cos A)(1 + \sec A)$ (A) $\sin A \tan A$ (B) $\sin A$ (C) $\cos A$ (D) $\sin A \cos A$ (E) $\tan A$
18. Solve completely the equations: $4x + y = 5$ and $2x^2 + y^2 = 17$. (A) $x = -2$ or $\frac{-2}{9}$ and $y = 3$ or $\frac{-37}{9}$ (B) $x = 2$ or $\frac{-2}{9}$ and $y = -3$ or $\frac{-37}{9}$ (C) $x = 2$ or $\frac{2}{9}$ and $y = -3$ or $\frac{37}{9}$ (D) $x = -3$ or $\frac{-37}{9}$ and $y = -3$ or $\frac{-37}{9}$ (E) $x = -2$ or 3 and $y = \frac{-2}{9}$ or $\frac{-37}{9}$
19. What is the value of $\det(A)$ if $A = \begin{pmatrix} 2 & 7 & 5 \\ 4 & 6 & 3 \\ 8 & 9 & 1 \end{pmatrix}$? (A) 38 (B) 140 (C) -60 (D) -42 (E) -38
20. If the sum to infinity of the series $x + x^3 + x^5 + \dots$ is $\frac{2}{3}$, Find the value(s) of x (A) $-2, \frac{1}{2}$ (B) $-3, \frac{1}{2}$ (C) 2, $-\frac{1}{2}$ (D) 1, 2 (E) $\frac{1}{2}, 2$
21. Eliminate θ from the equations: $x = 2 \cos \theta$ and $y = 3 \sin \theta$. (A) $9x^2 + 4y^2 = 36$ (B) $9x^2 + 4y^2 = 36$ (C) $9x^2 + 4y^2 = -36$ (D) $4y^2 - 9x^2 = 36$ (E) $x^2 + y^2 = 1$
22. If α and β are the roots of the quadratic equation $2x^2 + 3x + 7 = 0$, find the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (A) $-\frac{19}{14}$ (B) $\frac{15}{4}$ (C) $\frac{15}{4}$ (D) $-\frac{7}{2}$ (E) $-\frac{15}{4}$
23. Which of the trigonometrical ratios are positive for 960° ? (A) sine and tangent (B) tangent and cotangent (C) sine and cosine (D) secant and sine (E) cosecant and cotangent
24. For what values of m will the equation $(m+1)x^2 + 2(m+3)x + (2m+3) = 0$ have equal roots (A) 1 or -1 (B) -2 or 3 (C) 5 or -3 (D) 1 or 1 (E) 3 or 4
25. If the roots of the equation $2x^2 - 3x + r = 0$ are equal, find r . (A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $-\frac{2}{3}$ (D) $\frac{9}{8}$ (E) $\frac{1}{4}$
26. $-2 + 2i$ in polar form is given by (A) $\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$ (B) $\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$ (C) $2\sqrt{2} \left(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4}\right)$ (D) $2\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$ (E) $\left(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4}\right)$
27. Compute $\frac{2+9i}{5-2i}$. (A) $\frac{-8}{29} - \frac{49i}{29}$ (B) $\frac{-8}{29}$ (C) $\frac{49i}{29}$ (D) $\frac{-8}{29} + \frac{49i}{29}$ (E) $\frac{-49i}{29}$
28. If $n_{C_4} = n_{C_4}$, find n (A) 7 (B) 10 and 8 (C) 8 (D) 10 (E) -1 and 10
29. From question (28) above, find n_{C_2} (A) 28 (B) 45 (C) 21 (D) 73 (E) 88
30. If $\theta = \frac{3\pi}{6}$ in radians, what is θ in degrees? (A) 30° (B) 150° (C) 50° (D) 105° (E) 60°
31. Given that $\tan \theta = 1$ and $-360^\circ < \theta < 360^\circ$, what are the four possible values of θ (A) $315^\circ, 135^\circ, -45^\circ, -225^\circ$ (B) $-315^\circ, -135^\circ, 45^\circ, 225^\circ$ (C) $315^\circ, -135^\circ, -45^\circ, 225^\circ$

DEPARTMENT OF MATHEMATICS

FEDERAL UNIVERSITY, LAFIA

FIRST SEMESTER 2017/2018 EXAMINATION

Course Code: MTH 111 General Mathematics I (3 Units)

Instructions: Answer All Questions

Time: 2:30min

1. The first and the last term of an arithmetic progression are $7\frac{1}{2}$ and 93 respectively. If the common difference is $4\frac{1}{2}$. Find the sum of the series. (A) 1005 (B) 15 (C) 10005 (D) 10015 (E) 105
2. The number of radians in one complete revolution is (A) 3π (B) 2π (C) π (D) $\frac{\pi}{2}$ (E) 360°
3. If $A = \{5, 10, 15, 20, 25, 30, 35\}$ and $B = \{4, 6, 8, 10, 20, 30\}$, find $B \cap A$. (A) $\{4, 6, 8, 10, 20, 30\}$ (B) $\{5, 15, 25, 30\}$ (C) $\{4, 6, 8\}$ (D) $\{5, 15\}$ (E) non of the above
4. If there are 128 possible subsets in set X, compute the cardinality of set X. (A) -8 (B) 8 (C) 6 (D) 7 (E) 5
5. Solve $2\cos^2 x - \sin x = 1$. (A) $\sin x = \frac{\sqrt{3}}{2}$ or -1 (B) $\sin x = -\frac{1}{2}$ or 1 (C) $\sin x = 1$ or -1 (D) $\sin x = -\frac{1}{2}$ or $\frac{1}{2}$ (E) $\sin x = \frac{1}{2}$ or -1
6. Without solving the quadratic equation, define the nature of the roots of $4x^2 + x - 3 = 0$ (A) real and distinct (B) equal roots (C) not real roots (D) $-1, \frac{3}{4}$ (E) $2, \frac{3}{4}$
7. The compound angle identity $\cos(A+B)$ is given by (A) $\sin A \sin B + \cos A \cos B$ (B) $\cos A \cos B - \sin A \sin B$ (C) $\sin A \sin B - \cos A \cos B$ (D) $\sin A \cos B + \sin B \cos A$ (E) $\cos A \cos B + \sin A \sin B$
8. If $\cos \theta = \frac{\sqrt{3}}{2}$ and $\tan \theta$ is negative, what is $\sin \theta$? (A) $\sin \theta = \frac{1}{2}$ (B) $\sin \theta = -\frac{\sqrt{3}}{2}$ (C) $\sin \theta = -\frac{1}{2}$ (D) $\sin \theta = \frac{3}{2}$ (E) $\sin \theta = \frac{2}{3}$
9. The 2nd term of a geometry series is 24, the 5th term is 81. Find the first term. (A) 14 (B) $\frac{3}{2}$ (C) 12 (D) 16 (E) -12
10. Out of 6 women and 8 men, a committee consisting of two women and three men is to be formed. In how many ways can this be done if: one particular man must be on the committee? (A) 840ways (B) 525ways (C) 420ways (D) 300ways (E) 315ways
11. If the 21st and 22nd terms in the expansion of $(1+x)^{44}$ are equal. Find the value of x (A) 6 (B) 10 (C) $\frac{3}{2}$ (D) $\frac{5}{2}$ (E) $\frac{7}{8}$
12. In a geometric progression, the sum of the 2nd and 3rd term is 9 and the 7th term is eight times the 4th term. Find the common ratio. (A) -3 (B) $\frac{2}{3}$ (C) 3 (D) $\frac{3}{2}$ (E) 2

$$\frac{15}{2} - \frac{9}{2} \text{ or } 93 - \frac{15}{2} + \frac{9}{2}$$

$$\frac{6}{2} = 3$$

$$93 - 3 = 90$$

$$\frac{186 - 15 + 9}{2}$$

$$= \frac{180}{2} = 90$$

13. Find the range of values of p if the equation $px^2 - 4(p-1)x + 9 = 0$ has
 (A) $p \leq 4$ or $p \geq 2$ (B) $p \leq \frac{1}{2}$ or $p \geq 2$ (C) $p \leq \frac{1}{2}$ or $p \geq 3$ (D) $p \leq -\frac{1}{2}$
 (E) $p \leq \frac{1}{4}$ or $p \geq 4$

14. Given that $\begin{pmatrix} 3x & 5 \\ -1 & y \end{pmatrix} = \begin{pmatrix} 9 & 7 \\ 8 & -10 \end{pmatrix}$, what is the value of $3x + 2y^2$? (A) -10 (B) 29 (C) -209 (D) 209

15. If $\sin x = \frac{1}{2}$ and x is obtuse, what is $\cot x$? (A) $\sqrt{3}$ (B) $\sqrt{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) 3 (E) $-\sqrt{3}$

16. If the roots of the equation $2x^2 - 3x + r = 0$ are equal, find r . (A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $-\frac{2}{3}$ (D) $\frac{9}{8}$ (E) $\frac{1}{4}$

17. $-2 + 2i$ in polar form is given by (A) $\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$ (B) $\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$

- (C) $2\sqrt{2} \left(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4}\right)$ (D) $2\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$ (E) $\left(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4}\right)$

18. Compute $\frac{2+9i}{5-2i}$. (A) $\frac{-8}{29} - \frac{49i}{29}$ (B) $\frac{-8}{29}$ (C) $\frac{49i}{29}$ (D) $\frac{-8}{29} + \frac{49i}{29}$ (E) $\frac{-49i}{29}$

19. If $n_{C_4} = n_{C_4}$, find n (A) 7 (B) 10 and 8 (C) 8 (D) 10 (E) -1 and 10

20. From question (19) above, find n_{C_2} (A) 28 (B) 15 (C) 21 (D) 73 (E) 88

21. If $\theta = \frac{5\pi}{6}$ in radians, what is θ in degrees? (A) 30° (B) 150° (C) 50° (D) 105° (E) 60°

22. Given that $A = \begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$, compute $A^T A$. (A) $\begin{pmatrix} 25 & 2 \\ 2 & 5 \end{pmatrix}$ (B) $\begin{pmatrix} 25 & -2 \\ -2 & 5 \end{pmatrix}$ (C) $\begin{pmatrix} -25 & 2 \\ -2 & 5 \end{pmatrix}$ (D) $\begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix}$
 (E) $\begin{pmatrix} 10 & 5 \\ -2 & 5 \end{pmatrix}$

23. Given that $\tan \theta = 1$ and $-360^\circ < \theta < 360^\circ$, what are the four possible values of θ ?

- (A) $315^\circ, 135^\circ, -45^\circ, -225^\circ$ (B) $-315^\circ, -135^\circ, 45^\circ, 225^\circ$ (C) $315^\circ, -135^\circ, -45^\circ, 225^\circ$

- (D) $-315^\circ, 135^\circ, 45^\circ, -225^\circ$ (E) $-315^\circ, -135^\circ, -45^\circ, -225^\circ$

24. The real and the imaginary parts of $4 - i$ are r (A) 4 and 1 (B) -1 and 4 (C) 4 and -1 (D) 1 and 4
 (E) 4 and 0

25. Simplify $(\cos 2\theta + i \sin 2\theta)^2 (\cos \theta - i \sin \theta)^3$ using DeMoivre's theorem. (A) $(\cos 2\theta + i \sin 2\theta)$

- (B) $(\cos 2\theta - i \sin 2\theta)$ (C) $(\cos \theta + i \sin \theta)$ (D) $(\cos 2\theta - i \sin 2\theta)$ (E) $\cos 2\theta$

26. If $B = \begin{pmatrix} 5 & 1 \\ 3 & -2 \end{pmatrix}$, what is the value of $B^2 - 4B + 3I$? (A) $\begin{pmatrix} 5 & 1 \\ 3 & -2 \end{pmatrix}$ (B) $\begin{pmatrix} 11 & 1 \\ 3 & 8 \end{pmatrix}$ (C) $\begin{pmatrix} 11 & -1 \\ -3 & 18 \end{pmatrix}$
 (D) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (E) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

27. The descriptive notation for union of two sets A and B is (A) $A \cup B = \{x/x \in A \text{ and } x \in B\}$
 (B) $A \cup B = \{x: x \in A \cup x \in B\}$ (C) $A \cup B = \{x/x \in A \text{ or } x \in A\}$ (D) $A \cup B = \{x: x \in A \cap x \in B\}$
 (E) $A \cup B = \{x: x \in A < x \leq B\}$

28. In how many ways can 3 boys and 5 girls be arranged in a row so that all the three boys are together.
 (A) 4320 (B) 40320 (C) 4838400 (D) 2160 (E) 720.

29. Find the greatest common divisor of 24 and 30 (a) 9 (b) 12 (c) 6 (d) 8 (e) 4

30. The 7th term of an arithmetic progression is 27 and the 12th term is 47. Find the 5th term. (A) 8 (B) 20 (C) 19
 (D) 15 (E) 11

24 = 302

8

(E) $\sqrt{2}(\sqrt{3} - 1)$

- $$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_n = \frac{n}{2} [2 \times 3 + (n-1)4]$$

$$S_n = \frac{n}{2} [6 + 4n - 4]$$

$$S_n = \frac{n}{2} [6 - 4 + 4n]$$

$$\frac{2}{4} [2 + 4n]$$

$$\frac{n}{2} \cdot 2[1 + 2n]$$

$$n = [1 + 2n] \quad \text{or} \quad n = \sqrt{}$$